A STRATEGY FOR THE DROP OF $\frac{1}{16}$ AT ± 2

Setting as in `status.pdf`: $A_i = C(Y_2, P_i)$, $B_i = C(Y_{-2}, P_i)$, T the traceless complex of (S^3, K), mixed triangle $A_0 \xrightarrow{a} T \xrightarrow{b} B_1 \xrightarrow{c} A_0$ with homotopies g_A ($ba \simeq 0$), g_M ($cb \simeq 0$), g_B ($ac \simeq 0$). Levels (topological part): $a, b \le 0$; $c = \frac{3}{16}$; $g_A = -\frac{1}{8}$; $g_M = g_B = +\frac{1}{16}$. Everything below has a primed twin with the bundles exchanged.

1. What the ± 1 proof teaches

The map g_1 works because (i) its level is an index count, $-\frac{1}{8}$ for one parameter, (ii) its energy is positive on a simply connected cobordism, and (iii) it is *injective* because the middle term vanishes. At ± 2 the same family, g_A, has (i) and (ii) but not (iii): its boundary is ba. The only way to keep (i)–(ii) and recover (iii) is to restrict g_A to the chains that a kills.

2. The source

The map. $\ker a \subset A_0$ is a subcomplex, and

$$\Psi := g_A\big|_{\ker a} : \ \ker a \longrightarrow B_1$$

is a chain map: $dg_A x + g_A dx = ba\, x = 0$. Its level is $-\frac{1}{8} - \eta_g$, $\eta_g > 0$, for exactly the reasons of ± 1 (index -1, W' simply connected, no flat singular connection with both limits). It is injective on $H(\ker a)$ modulo the classes that bound in A_0 through a chain whose a-image is exact: triangle detection, $(g_A, b)\colon \mathrm{Cone}(a) \to B_1$ is a quasi-isomorphism, and Ψ is its restriction to $\ker a \subset \mathrm{Cone}(a)$. These are proved facts, not new ones.

What it gives. If the bottom class ξ of A_0 has a cycle representative in $\ker a$, then $m_1 \ge \frac{1}{8} + \eta_g > \frac{1}{8}$. This is the *non-absorbed* case. It covers the positive torus knots (parity forbids absorption) and it is where the data show $m \in (\frac{1}{8}, \frac{3}{16}]$.

3. The absorbed case and where $\frac{1}{16}$ lives

Suppose instead $a\xi = \tau \neq 0$ in $H(T)$. Write $\lambda = c(\xi)$, $\mu = c(\tau) \le \lambda - \eta_a$, and let $\kappa = \lambda + \delta$ be the lowest level of a class of $\ker a_*$; δ is the *depth of absorption*. Then Ψ applied at κ gives

$$m_1 \ \ge \ \tfrac{1}{8} - \delta + \eta_g \qquad \text{(Theorem 5.1 of } \texttt{strategy-exact}\text{)}.$$

So the whole problem is the single inequality

$$\delta \ \le \ \tfrac{1}{16} + \eta_g. \tag{1}$$

It is exactly tight in both equality cases. For $T(2,-3)$ in the primed triangle: $\lambda = -\frac{1}{24}$, $\kappa = \frac{1}{12}$, so $\delta = \frac{1}{8}$, and the g'_A-instanton on the first class of $\ker a'$ has $\eta_g = \frac{1}{16}$; for 4_1: $\delta = \frac{1}{10}$, $\eta_g = \frac{3}{80}$. In both, $\delta - \eta_g = \frac{1}{16}$, and $m_2 = \frac{1}{8} - \delta + \eta_g = \frac{1}{16}$. This is the structural reason for the two equalities, which `triangle-1` lacked. So (1) is not an artefact of the Seifert examples: it is the statement the examples saturate.

Where (1) *comes from.* The vertex relation at A_0 (the pentagon h_0, level $-\frac{1}{16}$, $b^+ = 1$, two parameters, γ_1 end) is

$$\mathrm{id}_{A_0} \ \simeq \ c\, g_A + g_M\, a.$$

Apply it to ξ: $\xi = [c\, g_A x + g_M \tau]$ with x a cycle for ξ. The right side has topological level $\lambda + \frac{1}{16}$ (both terms: $\frac{3}{16} - \frac{1}{8}$ and $\frac{1}{16} + 0$), minus energies. Now $w := c\, g_A x + g_M \tau$ is a cycle with $a_*[w] = \tau$, hence $\xi - [w] \in \ker a_*$, at level $\le \lambda + \frac{1}{16} - \min(\eta_g + \eta_c,\ \eta_a + E_M)$. If $\xi - [w] \neq 0$ this is (1) with room to spare. The one escape is

$$\xi = [w] = [\, g_M \tau + c\, g_A x \,] : \tag{2}$$

the bottom class is returned to itself by g_M from its own absorption image. A cosmetic pair needs (2) for both bottoms of both bundles, and then forces $\eta_a + E_M = \frac{1}{16}$ exactly on the returning instanton (the identity $c(\tau) + \frac{1}{16} - E_M = c(\xi)$ with $c(\tau) = c(\xi) - \eta_a$). The sixteenth is the width of this window: $\mathrm{level}(g_M) - \mathrm{level}(a)$.

4. The statement to prove, and why it is plausible

Conjecture R (no return). For a nontrivial knot with $\Delta_K = 1$, the bottom class of $I(Y_2, P_i)$ is not of the form (2): equivalently, g_M does not carry the bottom of T back onto the bottom of Y_2 at the chain level.

This is not an energy estimate. It is a statement about which generator g_M hits, i.e. about the return instanton: a one-parameter family on $(X^\circ_{-2}, D_r) \cup_{Y_{-2}} (-X')$ with the flat limit γ_1 at $L(4,1)$ and $b^+ = 1$. Reasons to believe it:

- *Zero-energy returns need common points.* A return with $\eta_a = 0$ is an absorption through a traceless representation with $\lambda \mapsto \pm 1$, binary dihedral at $(\frac{1}{4}, 0)$; there are $(\det K - 1)/2$ of these, so none when $\Delta_K = 1$. Both equality cases are of this type. This is the only mechanism the data ever show.

- *The return crosses $b^+ = 1$.* The family lives on $-X'$ glued to a singular handle, where the energy budget $\frac{1}{16}$ is the minimal odd cap energy $1/(4|S \cdot S|)$ of νS. A return of total energy exactly $\frac{1}{16}$ is a lens-end configuration: a flat singular connection plus the minimal cap. By the Thom principle we may assume Γ_K misses the four mixed points and is transverse to the lines; then no such flat connection exists, and (2) cannot be realised by a single instanton. What remains is a sum of several terms of which a cancellation would have to produce ξ; this is the genuinely open part.
- *Data.* In all 211 exact cases, (1) holds, equality only in the two binary-dihedral cases.

Test that decides it. Compute g_A, g_M and a at the chain level for the mirror torus knots and 4_1, in the pillowcase model (Section 6 of the uploaded paper gives the generators and levels; the maps are polygon counts with one corner at the mixed point). This checks (2) directly on the absorbed cases and shows what the returning instanton is. I have not done this.

5. **Summary**

Source: the one-parameter family g_A on W', restricted to $\ker a$. Injective and of level $-\frac{1}{8}$ for the same reasons as g_1. The drop it gives is $\frac{1}{8} - \delta + \eta_g$, and $\frac{1}{16}$ is a bound on the depth of absorption, forced by the vertex relation unless the bottom class is returned by g_M from the bottom of T (2). Conjecture R says it is not. The two equality cases are precisely returns at zero energy through binary dihedral representations, which do not exist for $\Delta_K = 1$.